Report - Version 15: Explicit Multi-Head Causal Self-Attention

Introduction

Version 15 preserves the Version 14 datasets, tokenizer, base training, checkpoints, held-out evaluation, instruction-tuning stage, and controlled generation. The only new concept is explicit multi-head causal self-attention.

The existing 16-dimensional attention space is split into two independent 8-dimensional heads. Q, K, V, and attention-output projection matrices keep the same shapes, so the model remains exactly 34,496 parameters. This isolates the multi-head computation without changing width, depth, optimizer, base-training budget, or SFT procedure.

```
Version 14 instruction-tuned pipeline
-> Q/K/V projected to 16 dimensions
-> split into 2 heads x 8 dimensions
-> independent causal attention in each head
-> concatenate head outputs to 16 dimensions
-> same 16 x 8 attention-output projection
-> preserved training + evaluation + SFT + generation
```

Goal of Version 15

- introduce two attention heads without increasing parameter count
- keep two Transformer blocks and context length 16 unchanged
- verify normalization and causal masking independently for every head
- preserve LAST/BEST base checkpoints and Version 13 evaluation
- preserve Dolly instruction tuning and response masking from Version 14
- close the V1-V15 arc with one isolated architectural evolution

Architecture and continuity

```
BPE + context 16
-> 8D token/position representation
-> Q/K/V: 8 x 16
-> reshape to 2 heads x 8D
-> per-head causal scaled dot-product attention
-> concatenate heads to 16D
-> output projection 16 x 8
-> residual + LayerNorm + FFN 8 -> 16 -> 8
-> second identical block + LM head
```

Continuity with Version 14

All data and training choices remain fixed: WikiText selected rows 10k/1k/1k, BPE 2,048, sequence caps 12k/2k/2k, batch size 32, learning rate 1.0, 20 base epochs, Dolly 256 selected rows split 192/32/32, instruction batch size 32, learning rate 0.15, and 8 SFT epochs.

Attention shapes

Element	Shape
Input representation	(batch, 16, 8)
Projected Q/K/V	(batch, 16, 16)
After split	(batch, 2, 16, 8)
Per-head attention	(batch, 2, 16, 16)
Concatenated heads	(batch, 16, 16)
Attention output	(batch, 16, 8)

Trainable parameters

Component	Parameters
Token embeddings	2048 x 8 = 16384
Positional embeddings	16 x 8 = 128
Transformer block 1	800
Transformer block 2	800
Output weights	8 x 2048 = 16384
Total	34496

Each Transformer block still contributes exactly 800 parameters:

Part	Shape	Params
Query projection	8 x 16	128
Key projection	8 x 16	128
Value projection	8 x 16	128
Attention output	16 x 8	128
LayerNorm 1 scale + shift	8 + 8	16
Feed-forward input	8 x 16	128
Feed-forward output	16 x 8	128
LayerNorm 2 scale + shift	8 + 8	16
Per block		800

Why the parameter count does not change

The Q, K, and V matrices remain 8 x 16 and the attention-output projection remains 16 x 8. Multi-head attention changes only how those 16 projected dimensions are reshaped and processed; it introduces no new trainable matrices.

Head	Dimension	Attention matrix
1	8	16 x 16
2	8	16 x 16

Explicit multi-head computation

After Q/K/V projection, tensors are reshaped from (batch, 16, 16) to (batch, 16, 2, 8) and transposed to (batch, 2, 16, 8). Each head independently computes scaled scores, applies the causal mask, normalizes with softmax, and combines values.

```
Q/K/V projected: (B,16,16)
-> split_heads: (B,2,16,8)
-> scores: (B,2,16,16)
-> causal softmax independently per head
-> head outputs: (B,2,16,8)
-> transpose + concatenate: (B,16,16)
-> output projection: (B,16,8)
```

Preserved training pipeline

Base training still uses the same clean interruption at epoch 10, LAST restoration, resume through epoch 20, and BEST model selection. WikiText held-out evaluation from Version 13 and Dolly SFT from Version 14 are preserved, so the architectural change remains isolated.

Properties verified by the notebook

- NUM_HEADS = 2 and HEAD_DIM = 8
- attention tensor from each block has shape (1, 2, 16, 16) for the inspected example
- every attention row in every head sums to 1
- all future-position entries are zero in every head
- total parameter count remains 34,496

Preserved evaluation, SFT, and generation

Version 15 retains WikiText cross-entropy, perplexity, top-1/top-5 token accuracy, base BEST restoration, the deterministic Dolly 192/32/32 split, response-focused masked SFT, BEST instruction selection, and the base-versus-tuned controlled generation procedure.

Instruction-stage facts

Property	Value
Prompt format	Instruction / optional Context / Response
Response mask	0 on prompt targets, 1 on response targets
Instruction epochs	8
Instruction learning rate	0.15
Instruction generation	greedy and top-p 0.9, 16 new BPE tokens
Tokenizer	same BPE; not retrained on Dolly

Included checks

- two heads and head dimension 8
- per-head attention shape, normalization, and causal masking
- 23 trainable variables and 34,496 parameters
- base LAST/BEST resume behavior remains valid
- WikiText metric consistency is preserved
- Dolly train/validation/test source rows remain disjoint
- response masks remain binary and response-focused
- instruction tuning starts from base BEST and updates parameters
- BEST instruction checkpoint restores correctly
- controlled generation remains reproducible
- the final test cell reports All checks passed

What Version 15 teaches

Version 15 makes the conceptual difference between one attention space and multiple attention heads visible without confusing that lesson with a larger model. The same Q/K/V projection width is reorganized into two independently normalized causal attention spaces and then recombined.

```
Version 14 pipeline
-> Q/K/V 16D
-> split 2 x 8D
-> causal attention per head
-> concatenate
-> same output projection
-> complete pipeline preserved
```

Limitations

- two heads are still a minimal multi-head configuration
- embedding/model width remains 8 dimensions
- context length remains fixed at 16 tokens
- no dropout, Adam, learning-rate schedule, or gradient clipping is added
- depth and width are not increased
- instruction tuning remains small and does not include preference optimization or RLHF

Conclusion

Version 15 closes the current V1-V15 arc with explicit two-head causal self-attention while preserving parameter count, training, evaluation, and instruction tuning. The final architectural change is isolated, directly inspectable, and consistent with the project philosophy of one version, one concept, one verifiable result.

Future versions will introduce new components and gradually evolve the architecture.